

Master Technical Architecture & Implementation Blueprint

Amazon ML Challenge 2026 — Business Entity Resolution

Canonical editable source. Generated 2026-09-26. Single source of truth for implementation. Derived strictly from (a) the official problem statement (`ProblemStatement.txt` / `CHALLENGE.md`), (b) the current repository state, and (c) the **frozen** candidate-generation design (`evidence_source_50_50_heavy100_v1`). No organizer requirement, dataset fact, benchmark number, or repository component is invented; every figure is traceable to a cited artifact. Where the repository conflicts with the brief, the discrepancy is reported (see §11, §32), not silently resolved.

Labeling legend

Every architectural element is tagged:

- **[ORG]** *Organizer Requirement* — mandated by the problem statement / validator.
- **[PDD]** *Project Design Decision* — a deliberate choice by this project.
- **[ENG]** *Engineering Constraint* — forced by environment (RAM, OS, licenses).
- **[EXP]** *Experimental Decision* — chosen because a logged experiment supports it.
- **[FUT]** *Future/Optional Enhancement* — planned/optional; not yet built.

Build-status legend

-  **Existing** — code + artifact present and verified.  **In progress** — scaffold/contract only, computation not implemented.  **Planned** — described in plan docs only; no code yet.

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1. Executive Summary

This project resolves business entities across **three noisy sources**. **Source 1 (S1) is the deduplicated reference [ORG]** — for every S1 entity the system must return all matching S2 and/or S3 records (zero, one, or many). Two graded TSVs are produced: `output/matching_results.tsv` (final matches — the leaderboard file) and `output/candidate_pairs.tsv` (the *exact* candidate set fed to the matcher, the last blocking stage). **Both count toward ranking; every matched ID must also appear in candidates; smaller per-S1 candidate sets rank higher. [ORG]** The optimization target is **F_{0.5} computed per S1 then macro-averaged** — precision-heavy (a false merge costs $\approx 2 \times$ a miss); a correctly predicted singleton scores exactly **1.0. [ORG]**

The architecture is a disciplined **retrieve-then-score** pipeline with a hard internal boundary: a **frozen, label-free candidate generation** stage (an immutable upstream contract) feeds a **pairwise tabular matcher** and an **S1-level decision policy** that assembles per-entity match sets. Candidate generation is deliberately separated from candidate scoring, which is separated from entity-level decision-making — three distinct concerns with three distinct evaluation layers.

Built & verified today (✓): deterministic **normalization** (`normalize.py` , frozen SHA `b3508b60...`); **evaluator** (`evaluate.py` , frozen SHA `e99d7f0a...`) implementing macro-F_{0.5} exactly (spec example reproduces **0.714**); the **candidate-development split** (seed 42, val 220,531) and the **matcher split v1** leakage firewall (`model_train` 1,765,608 / `model_calibration` 110,341 / `model_final_eval` 110,341 / `candidate_dev` 220,531; cross-split S1 overlap **0**, labelled-target overlap **0**); the **FROZEN candidate policy** `evidence_source_50_50_heavy100_v1` — **34,568,979** candidates over 16 hash partitions, pair recall **0.878084** (recovered **670,785 / 763,919** dev GT links), candidates/S1 mean **156.75** / p99 **213** / max **292**, reproduced with **0 added / 0 removed / 0 duplicate** identities (freeze gate **passed**, manifest checksum `221fef24...9465b067`); the **candidate oracle ceiling** of macro-F_{0.5} $\approx$ **0.9486**; and a **65-feature contract**.

Pilot executed, full-scale pending (●): a **pilot/smoke end-to-end has run** — a 65-feature matrix computed on **1,098,300 pilot candidates** (audit 0/0/0), three checksummed negative-sampling policies (`hybrid_source_balanced` selected), and **Deterministic / Logistic / LightGBM-smoke matchers fit and thresholded** (pilot macro-F_{0.5} 0.628979 / 0.781286 / 0.851852 — **pilot/calibration numbers, not leaderboard, not L4**). Full-scale feature materialization over the model-development splits is **actively running** (`model_development_features/`). This corrects an earlier "scaffold/no `.fit` " draft — see the discrepancy log (§36).

Planned only (🟡): full-scale GBM bake-off + hyperparameter search, calibration & threshold tuning on the full split, singleton & multi-match decision policy, France zero-shot probes, **test inference**, and the two output TSVs (`output/` is currently **empty**). `model_final_eval` remains **firewalled and untouched** — **no end-to-end L4 score exists** and none is claimed.

Governance [ORG]: strictly **no external business-identity lookup, enrichment, geocoding, registry, or APIs** (disqualifying); the frozen policy asserts `label_use = all false`; the final model must be **MIT/Apache-2.0, ≤8B params** — satisfied by a tabular GBM, **not** an LLM.

2. Challenge Overview

The Amazon ML Challenge 2026 Business Entity Resolution task provides three record sources describing businesses, plus a ground-truth mapping on the training partition. **[ORG]** The sources are noisy and heterogeneous: the *same* real-world business may appear with transliterated names (Devanagari→Latin in India), accented forms (France), legal-suffix variants (Pvt/Private, Ltd/Limited, Corp/Corporation, Sarl/SCI), `&` ↔ `and` and substitutions, word-order swaps, typos, leading junk (`--` , `<<`), or domains-as-names. Roughly 3% of S2/S3 addresses are empty. **[ORG]**

S1 is authoritative and deduplicated [ORG]. The deliverable is, for every S1 entity, the set of S2/S3 records that denote the same business. This is fundamentally **one-to-many** entity resolution, not record deduplication and not 1:1 linkage: on training data an S1 entity has on average **3.46** matches (max **11**), **80.5%** match both an S2 and an S3, and **76.8%** match multiple records within a single source. **5.6%** of S1 are true **singletons** (no match at all).

Two outputs, both graded [ORG]. Unusually, the *candidate set itself* is a scored deliverable: `candidate_pairs.tsv` must be the exact final blocking output handed to the matcher, and tighter per-S1 candidate sets rank higher. This couples the blocking and matching stages: the system cannot "over-retrieve then hope the matcher cleans up" without paying a candidate-size penalty.

Open-set countries [ORG]. Training covers only **US** and **India**; the **test set adds France** (~15% of test S1, zero training rows). The architecture must therefore be country-agnostic and generalize zero-shot to an unseen country.

Strict fair-play [ORG]. Learning must come only from the provided data. Any external business-identity lookup, enrichment, commercial ER API, government/business registry, geocoding API, or internet-based entity enrichment is **disqualifying**. Deterministic local normalization (stdlib `unicodedata`) is permitted; looking up a real business is not.

Model & resource envelope [ORG/ENG]. The final model must be MIT/Apache-2.0 licensed and ≤8B parameters; an LLM is explicitly *not* the default. The working environment is memory-bound (~0.8 GB planning budget of free RAM), mandating out-of-core execution at the scale of millions of records and hundreds of millions of candidate pairs (\$19).

3. Official Requirements

Enumerated verbatim-in-intent from `ProblemStatement.txt` / `CHALLENGE.md` / `validate_submission.py`. All rows are **[ORG]**.

Task & matching semantics

- O1. Three sources (S1/S2/S3); **S1 is the deduplicated reference**.
- O2. For each S1 entity, output all matching S2/S3 records — **zero, one, or many**.
- O3. Matches may span both S2 and S3 simultaneously (one-to-many, cross-source).

- O4. **Singletons** exist: an S1 entity may legitimately match nothing.

Data & schema

- O5. Records have `entity_id`, `business_name`, `business_address`, `country`; TSV format.
- O6. ID conventions: `S1-*`, `S2-*`, `S3-*` prefixes; S2/S3 namespaces disjoint.
- O7. Ground truth (train only): `source1_entity_id`, `matched_entity_ids` (comma-joined; empty = singleton).
- O8. Data is noisy (transliteration, accents, suffixes, typos, junk, domains-as-names, ~3% empty addresses).
- O9. **Open-set country**: test adds France (0 training rows).

Outputs & validation

- O10. `matching_results.tsv` — headers exactly `source1_entity_id→matched_entity_ids`; one row per test S1; empty field = singleton.
- O11. `candidate_pairs.tsv` — headers exactly `source1_entity_id→candidate_entity_ids`; the **exact final candidate set** fed to the matcher.
- O12. `candidate_pairs.tsv` **counts toward ranking; smaller per-S1 candidate sets rank higher**.
- O13. **Every matched ID must appear in `candidate_pairs.tsv`** (matches $\subseteq$ candidates).
- O14. Real TABs; **IDs and S1 keys are NOT stripped** → no stray spaces, no spaces after commas.
- O15. Lists contain only S2-/S3- IDs; no self-matches to S1-; no duplicate IDs in a list; no duplicate S1 rows; every test S1 present; no IDs absent from the test set.
- O16. Validation gate: `validate_submission.py --matching ... --candidate ... --test-dir ...` must print `PASS`.

Metric

- O17. $F_{0.5} = (1.25 \cdot P \cdot R) / (0.25 \cdot P + R)$, **per S1 then macro-averaged**.
- O18. Empty truth + empty prediction = **1.0**; any prediction on a true singleton = **0.0**; $\beta=0.5 \Rightarrow$ precision $\approx 2 \times$ recall.
- O19. Candidate recall is a diagnostic, **not** the final leaderboard score.

Model, resources, fair-play

- O20. Final model **MIT/Apache-2.0**, **$\leq 8B$ params**; LLM not the default.
- O21. Solution must **scale** to millions of records / hundreds of millions of pairs.
- O22. **No external identity lookup / enrichment / registry / geocoding / commercial ER APIs** (disqualifying).
- O23. Reproducible from `code/` alone; pinned versions; fixed seeds.

4. Requirement-to-Architecture Traceability

Every explicit organizer requirement (§3) maps to an architectural consequence, an implementation component, and a verification mechanism. Status is in §30/§31.

Organizer Requirement	Architectural Consequence	Implementation Component	Verification
O1 S1 = reference	Pipeline keyed on S1; one output row per S1	<code>split.py</code> , submission assembler	<code>integrity_report.md</code> : 1 GT row/S1
O2/O3 zero..many, cross-source	Per-candidate keep/drop; no top-1	matcher + S1 decision policy (§16–17)	oracle all/some/none recovered counts
O4 singletons	Empty-prediction path; conservative τ	decision policy (§17)	evaluator empty/empty=1.0
O5–O7 schema/IDs/GT	Typed columnar store; id namespaces	<code>io.py</code> , <code>normalize.py</code>	<code>integrity_check.py</code> (uniqueness, disjoint)
O8 noise	Multi-key deterministic normalization	<code>normalize.py</code> (frozen)	unit tests; baseline recall lift 0.47→0.88
O9 open-set country	country used only as <i>agreement</i> ; LOCO probes	feature spec; §25	<code>country_open_set:true</code> ; France path ()
O10/O11 two TSVs	Assembler emits both; candidate TSV = frozen artifact	submission () + <code>final_candidate_policy_parts</code>	<code>validate_submission.py</code> → PASS
O12 smaller sets rank higher	source 50/50 + heavy-100 cap	frozen policy v1 (§11)	p99 755→213, max 1537→292
O13 matches $\subseteq$ candidates	Decision policy only <i>filters</i> candidates	§17 invariant	validator subset check
O14/O15 format rules	Exact-format writer; no strip	assembler; validator	validator FAIL on malformed
O16 PASS gate	<code>validate_submission.py</code> is release gate	<code>utils/validate_submission.py</code> (✓)	manual gate run
O17/O18 macro- $F_{0.5}$	Per-S1 P/R/ $F_{0.5}$ → macro	<code>evaluate.py</code> (frozen ✓)	spec example → 0.714
O19 recall $\neq$ score	4 distinct eval layers	oracle (§12), §18	recall 0.878 vs oracle 0.9486 kept separate
O20 $\leq 8B$, no LLM	Tabular GBM plan	<code>matcher/</code> ()	model card at selection
O21 scale	Out-of-core, 16 hash partitions, DuckDB spill	<code>materialize.py</code> (✓)	RSS ≈ 648.5 MiB reproduction, spill 0
O22 fair-play	<code>label_use=false</code> ; stdlib-only normalize	frozen policy; <code>normalize.py</code>	policy manifest assertions
O23 reproducible	SHA-256 manifests, seeds, pinned reqs	manifests, <code>requirements.txt</code>	freeze gate 0/0/0 identity diff

Diagram D1 — Overall system architecture (responsibilities annotated; full set in `work/amazon_ml_challenge_architecture_diagrams.md`):

```

flowchart TD
    DS["Challenge datasets<br/>S1 / S2 / S3 (+GT train only) [ORG]"] --> IV["Data ingestion +<br/>typed columnar store, integrity checks"]
    IV --> NR["Normalization + representations<br/>frozen normalize.py ✓"]
    NR --> CG["Frozen candidate generation v1<br/>4 label-free passes ✓"]
    CG --> CA["Candidate artifacts<br/>16 hash partitions, checksummed ✓"]
    CA --> FE["Pair feature extraction<br/>65-d pilot done, full-scale 🟡"]
    FE --> CM["Candidate matcher<br/>tabular GBM (smoke fit) 🟡"]
    CM --> DP["S1-level decision policy<br/>threshold + singleton + multi-match ☐"]
    DP --> MR["matching_results.tsv"]
    CA --> CP["candidate_pairs.tsv (exact set)"]
    MR --> VAL["Official validator<br/>validate_submission.py ✓"]
    CP --> VAL
  
```

5. Problem Definition

Formal statement [ORG]. Given three record sets — S1 (deduplicated reference), S2, and S3 — each record carrying `entity_id`, `business_name`, `business_address`, `country`, produce for every S1 entity e the set $M(e) \subseteq (S2 \cup S3)$ of records denoting the same real-world business. $M(e)$ may be empty (singleton), a single record, or many records spanning both S2 and S3. This is **one-to-many, cross-source entity resolution** — not deduplication (S1 is already deduplicated) and not 1:1 linkage (mean **3.46** matches/S1, max **11**; `candidate_refinement_report.md`, `integrity_report.md`).

Objective [ORG]. Maximize $\text{macro-F}_{0.5} = \text{mean over S1 of } F_{0.5}(e)$, where per entity $F_{0.5}(e) = (1.25 \cdot P(e) \cdot R(e)) / (0.25 \cdot P(e) + R(e))$, $P(e) = |\text{pred} \cap \text{truth}| / |\text{pred}|$, $R(e) = |\text{pred} \cap \text{truth}| / |\text{truth}|$. Edge cases fixed by spec: **empty truth $\wedge$ empty pred = 1.0**; **any non-empty pred on empty truth = 0.0**; $\beta=0.5 \Rightarrow$ a false merge costs $\approx 2\times$ a miss. The spec worked example evaluates to **0.714** (reproduced by frozen `evaluate.py`, §18).

Why the metric shapes the whole design [PDD]. Because precision dominates and a true singleton is worth a full 1.0 only when left empty, the system is engineered to be **conservative at the decision boundary** (§17) and to treat the singleton path as first-class, not an afterthought. Because `candidate_pairs.tsv` is *also* graded on tightness (O12), the blocking stage cannot over-retrieve to protect recall without a size penalty — recall and candidate-size are co-optimized, not traded freely.

Decomposition [PDD]. The problem factors into three separable sub-problems, each with its own evaluation layer (§12, §18):

1. **Retrieval / blocking** — bound $M(e)$ inside a small candidate set $C(e)$; measured by *pair recall* and *candidate size*. (FROZEN, §11.)
2. **Pairwise scoring** — for each (e, c) , $c \in C(e)$, estimate $P(\text{match})$; measured by pair precision/recall on held-out labels. (🟡 piloted — smoke matcher fit; full-scale 🟡)
3. **Entity decision** — turn per-pair scores into $M(e)$ under the $F_{0.5}$ objective, including the singleton call; measured by end-to-end $\text{macro-F}_{0.5}$. (🟡)

Hard invariant [ORG]: $M(e) \subseteq C(e)$ for every e (O13). The decision stage may only *filter* candidates; it can never introduce an ID absent from `candidate_pairs.tsv`. This makes retrieval pair recall (**0.878084**, §11) a **hard upper bound** on achievable recall, and the candidate oracle ($\text{macro-F}_{0.5} \approx \mathbf{0.9486}$, §12) a **hard upper bound** on achievable score for this frozen candidate set.

6. System Goals and Non-Goals

Goals [PDD unless tagged].

- **G1.** Maximize held-out macro- $F_{0.5}$ under the precision-heavy $\beta=0.5$ regime [**ORG objective**].
- **G2.** Emit two validator-clean TSVs where `matches` $\subseteq$ `candidates` and `candidate_pairs.tsv` is the *exact* set the matcher scored [**ORG O10–O16**].
- **G3.** Keep per-S1 candidate sets tight (smaller ranks higher) without sacrificing the frozen recall ceiling [**ORG O12**] — achieved p99 **213**, max **292** (§11).
- **G4.** Generalize zero-shot to **France** (unseen country) — country enters only as an *agreement* signal, never one-hot [**ORG O9**].
- **G5.** Run end-to-end within a **~0.8 GB free-RAM** planning budget at 2M+ S1 / 5M+ S2 / 5M+ S3 scale via out-of-core execution [**ENG**] (measured peaks §19).
- **G6.** Be fully reproducible from `code/` with pinned versions, fixed seeds, and SHA-256 manifests [**ORG O23**].
- **G7.** Preserve a hard **leakage firewall** between matcher training and evaluation splits [**PDD**].

Non-Goals.

- **N1.** No external business-identity lookup, enrichment, geocoding, registry, or commercial ER API — **disqualifying** [**ORG O22**]. The frozen policy asserts `label_use = all false`.
- **N2.** No LLM as the final matcher; the model envelope is MIT/Apache-2.0, $\leq 8B$ params, satisfied by a tabular GBM [**ORG O20**].
- **N3.** No top-1 / 1:1 assignment — the task is one-to-many; forcing a single best match is structurally wrong (76.8% match multiple within one source) [**ORG O3, data**].
- **N4.** No redesign or regeneration of candidate policy v1 — it is a **frozen upstream contract** (§11) [**PDD**].
- **N5.** No optimization against a public leaderboard signal; trust the held-out `model_final_eval` split [**PDD**].
- **N6.** No hard-coding of the {US, India} country set; treat country as an open set [**ORG O9**].

Design tension explicitly accepted [PDD]. The frozen candidate recall of **0.878084** caps recall: ~12.2% of true links (**93,134** of 763,919 dev GT links) are unreachable by any downstream matcher. This is a deliberate recall-for-precision-and-size trade (§11–§12): the $F_{0.5}$ objective rewards it, and the oracle shows the ceiling is still **0.9486**, far above the recall figure — recall loss does not translate 1:1 into score loss because most missed links are on entities that still match other records.

7. Data Architecture

Sources and scale [ORG facts, profiled]. Two disjoint partitions — **train** (labelled) and **test** (unlabelled) — each with three record sets (`integrity_report.md`):

Partition	S1	S2	S3	Countries
train	2,206,821	5,034,616	5,285,603	{US, India}
test	1,732,544	4,887,273	5,082,316	{US, India, France }

Test S1 country breakdown: India **809,986** / US **663,106** / France **259,452** (~15% France, **zero training rows** — the open-set challenge, O9). Ground truth exists **train-only**: 7,638,365 matched pairs, avg **3.461**/S1, max **11**, singletons **123,247 (5.6%)**; target rows S3- **3,944,746** / S2- **3,693,619**.

Storage strategy [ENG/PDD]. The dominant constraint is **~0.8 GB free RAM** against ~5M-row string tables. Therefore:

- Ingest TSV → **typed columnar Parquet** once; never `pd.read_csv` a full 5M-row string table into RAM.
- **Integer-code entity IDs** for joins; keep the string ID only for final emission.
- Big joins run in **DuckDB** with a bounded `memory_limit` and a `temp_directory` so hash joins/sorts **spill to disk**; scans use **polars streaming/lazy**.
- Candidate materialization is **partitioned into 16 shards** by `hash(source1_entity_id) % 16` so no stage holds the full 34.6M-row candidate table in memory.

Data integrity guarantees [verified — `integrity_check.py` , `integrity_report.md`].

- `entity_id` unique within every file; S2/S3 id namespaces **disjoint**.
- Every GT id resolves in-source (**0 missing**); **0** duplicate S1 GT rows; **0** intra-list dup ids.
- **Each S2/S3 record matches ≤1 S1** (cross-S1 sharing = **0**) → splitting on S1 partitions the labels with **no target leakage** (the firewall in §13 depends on this fact).
- `business_name` never empty (0); empty `business_address` S2 ≈ **3.36%** / S3 ≈ **3.33%** (the ~3% in O8).

Diagram D2 — Data ingestion and integrity flow:

Diagram source 2 (Mermaid — render via `mermaid-cli` for graphics)

```
flowchart LR
    RAW["Raw TSV<br/>train + test (S1/S2/S3)"] --> ING["Ingest once<br/>typed columnar Parquet"]
    ING --> IDC["Integer-code IDs<br/>keep string id for emit"]
    IDC --> INT["Integrity checks<br/>uniqueness, disjoint ns, GT resolves"]
    INT -->| "0 missing / 0 dup / ≤1 S1 per target" | OK["Verified store<br/>DuckDB + Parquet"]
    INT -->| violation | STOP["Halt: report, do not proceed"]
```

8. Data Schemas

Record schema [ORG O5]. All six input files share one schema; TSV with a header row.

Column	Type	Notes
<code>entity_id</code>	string	Prefixed <code>S1-</code> / <code>S2-</code> / <code>S3-</code> (O6); unique within file; not stripped by validator
<code>business_name</code>	string	Never empty (verified); noisy: translit, accents, suffixes, junk, domains
<code>business_address</code>	string	~3% empty in S2/S3; free-form
<code>country</code>	string	Open set; train {US, India}, test adds France

Ground-truth schema [ORG O7] (train only).

Column	Type	Notes
<code>source1_entity_id</code>	string	One row per train S1; the reference key
<code>matched_entity_ids</code>	string	Comma-joined S2-/S3- ids; empty ⇒ singleton

Output schemas [ORG O10/O11] — validator-exact.

File	Header (real TABs)	Row semantics
<code>output/matching_results.tsv</code>	<code>source1_entity_id→matched_entity_ids</code>	one row per test S1; empty field = singleton
<code>output/candidate_pairs.tsv</code>	<code>source1_entity_id→candidate_entity_ids</code>	the exact last-stage candidate set fed to the matcher

Format invariants the validator enforces [ORG O14/O15] (breaking any = rejection):

- Real TAB field separator; comma-joined lists with **no space after commas**.
- S1 key and IDs are **not stripped** → emit **no** leading/trailing whitespace.
- Lists contain only S2-/S3- ids; **no** S1 self-matches; **no** duplicate id in a list; **no** duplicate S1 rows; **every** test S1 present; **no** id absent from the test set.

Internal schemas [PDD] (not organizer-facing).

- **Candidate artifact** (frozen): (source1_entity_id, candidate_entity_id, provenance_bitmask, evidence_score) across 16 hash-partitioned Parquet parts; **34,568,979** rows total (§11).
- **Feature matrix** (🟡 contract only, §14): (pair_key, 65 numeric features, [train-only label]), float32-compact.
- **Split table** (✅): split_s1.parquet (dev split) + matcher-split assignment (§13).

9. Normalization Architecture

Status: ✅ **Existing, FROZEN.** code/.../normalize.py, SHA-256 b3508b60... (pinned in final_candidate_policy_manifest.json as a frozen input). Because candidate generation is frozen on top of it, normalization is itself immutable — any change would invalidate the candidate artifact's identity checksum (§11).

Purpose [PDD]. Collapse the O8 noise classes into stable comparison keys **using only deterministic, local, stdlib algorithms** — satisfying fair-play (O22): normalization *transforms* text, it never *looks up* a business.

Transform pipeline [PDD/EXP]. Applied to business_name and business_address:

1. **Unicode fold** — unicodedata NFKD + accent stripping (France accents; O8). Stdlib only.
2. **Transliteration bridge** — Devanagari↔Latin handling so India records collide on a common key (O8).
3. **Case / whitespace** — lowercase, collapse runs, strip leading junk (-- , <<), trim.
4. **Punctuation & connective canon** — & ↔ and , drop non-alphanumerics to spaces.
5. **Legal-suffix canon** — Pvt/Private, Ltd/Limited, Corp/Corporation, Sarl/SCI, etc. → a canonical token or removal, producing a **name-without-suffix** key.
6. **Domain-as-name** — strip URL/domain scaffolding to the core token.
7. **Token & sorted-token keys** — token set, and a **sorted-name** key (word-order-swap invariant).

Derived keys consumed downstream [PDD]. country (agreement only), name_nosuffix, sorted_name, name_token set, address_token set, exact_address. These are exactly the keys the four frozen retrieval passes block on (§10).

Evidence it works [EXP — baseline_blocking.md]. Exact (country, name_nosuffix) blocking recovers only **47.02%** of val true pairs (US 49%, India 44%); the sorted-key variant **50.61%**. Exact-name blocking is **provably insufficient** — this measured gap is the justification for the four-pass fuzzy / token / address design (§10), which lifts pair recall to **0.878** (§11). The jump **0.47 → 0.88** is the normalization+retrieval architecture's headline validated result, not an assumption.

Governance note [ORG]. Nothing here contacts a network, reads an external dataset, or resolves a real business. All transforms are pure functions of the input string + Python stdlib tables.

10. Candidate Generation Architecture

Status: ✅ **Existing, FROZEN (design described here; policy numbers in §11).** Label-free by construction — label_use = all false in the policy manifest, satisfying O22.

Four retrieval passes [EXP — candidate_experiment_summary.md]. Each pass blocks S2US3 targets to each S1 on a different key, then results are unioned and deduplicated. Single-pass val recall:

Pass	Key	Provenance bit	Single-pass recall
P1 sorted_name	word-order-invariant name key	1	0.5061
P2 exact_address	normalized full address	2	0.0832
P3 name_token	shared name tokens	4	0.5317
P4 address_token	shared address tokens	8	0.7965

Provenance bitmask [PDD]. Each surviving (S1, target) pair carries an integer bitmask = OR of the passes that produced it (sorted_name=1, exact_address=2, name_token=4, address_token=8). Dedup is a bitwise OR over passes, so provenance is preserved without row duplication — and becomes a **feature group** for the matcher (§14) at zero extra cost.

IDF-aware evidence ranking [EXP/PDD]. Token passes over-generate on frequent tokens. Each candidate gets $evidence_score = \sum_{\{shared\ tokens\ t\}} 1 / target_df(t)$ (rare shared tokens weigh more). This score drives per-S1 pruning (§11). Ranker pinned as evidence_ranker_v1.json (SHA a0fb606f...); rank shards rank_shards (SHA e2db9fd4..., 106,487,165 rows, 128 parts). Cumulative unioned recall before pruning reaches **0.9425**; the per-pass-capped canonical union sits at **0.8614** (658,063 recovered) — the *intermediate* figure, distinct from the frozen-final **0.878084** (§11, §33 discrepancy note).

Diagram D3 — Candidate generation (four passes → union → rank):

Diagram source 3 (Mermaid — render via mermaid-cli for graphics)

```
flowchart TD
    N["Normalized S1 + S2/S3 keys<br/>frozen normalize.py ✓"] --> P1["P1 sorted_name<br/>bit 1"]
    N --> P2["P2 exact_address<br/>bit 2"]
    N --> P3["P3 name_token<br/>bit 4"]
    N --> P4["P4 address_token<br/>bit 8"]
    P1 --> U["Union + dedup<br/>provenance = bitwise OR"]
    P2 --> U
    P3 --> U
    P4 --> U
    U --> R["IDF evidence rank<br/>score = Σ 1/target_df(token)"]
    R --> PR["Per-S1 pruning + quotas<br/>(frozen policy v1, §11)"]
```

11. Frozen Candidate Policy v1

Status: ✓ Existing, FROZEN — immutable upstream contract. Policy id evidence_source_50_50_heavy100_v1 ; policy SHA 46fd324b...; freeze gate passed=true (final_candidate_policy_manifest.json , final_candidate_artifact_manifest.json , candidate_refinement_report.md). **This section is descriptive only. Do NOT redesign, re-tune, or regenerate it** (§28, §36) — doing so breaks every downstream identity checksum.

Selection [EXP]. 13 candidate policies were compared in candidate_refinement_report.md ; the selected final policy applies, on top of the ranked four-pass union (§10):

- **Source 50/50 quota** — reserve balanced capacity for S2 and S3 targets per S1, so one noisy source cannot crowd out the other.
- **Heavy sorted-name cap 100** — cap the sorted-name pass at 100 for "heavy" S1 (blocks ≥120 records), taming quadratic blow-up on common names while protecting recall.

Frozen artifact — the graded candidate_pairs.tsv source:

Property	Value
Candidate rows	34,568,979
Partitions	16 by <code>hash(source1_entity_id) % 16 ; partition_manifest_sha256 221fef24...9465b067</code>
Artifact bytes	217,174,342
Recovered GT links (dev)	670,785 / 763,919
Pair recall (dev)	0.878084
S2 recall / S3 recall	0.881595 / 0.874805
India recall / US recall	0.875810 / 0.879595
Both-address-present / either-missing	0.884902 / 0.729202
Candidates/S1 mean	156.75
Candidates/S1 p99 / max	213 / 292 (from uncapped p99 755 / max 1537)
Zero-candidate S1s	230
Identity reproduction	0 added / 0 removed / 0 duplicate , equal=true

Recall discrepancy — reconciled, not hidden [per §36]. Two recall numbers appear in the evidence and are **both correct for different objects**: **0.8614** (658,063 recovered) is the *intermediate* "canonical IDF-100 per-pass-capped union" (`candidate_experiment_summary.md`); **0.878084** (670,785 recovered) is the *selected final frozen policy* `evidence_source_50_50_heavy100_v1` (`candidate_refinement_report.md`). The frozen-final number supersedes the intermediate; they are not in conflict.

Recall ceiling consequence [ORG O13]. Because `matches` $\subseteq$ `candidates` , **0.878084** is a hard recall ceiling and the **93,134** unreached links (mostly on either-address-missing pairs, recall 0.729) are permanently out of reach for this policy — a bounded, measured, accepted loss (§6, §12, §32).

12. Candidate Oracle and Quality Ceiling

Status:  **Existing** (`candidate_oracle_report.md`). The oracle answers one question: *if the matcher were perfect on the frozen candidate set, what macro-F_{0.5} is achievable?* It labels each candidate with the ground truth and computes the metric — an **upper bound**, not a live score (O19).

Ceiling for frozen policy v1:

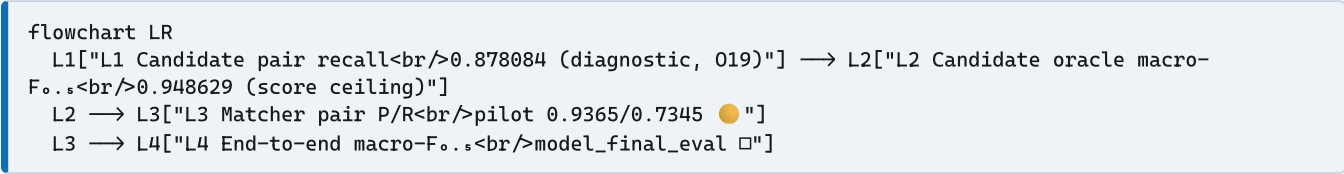
Metric	Value
Oracle macro-F_{0.5}	0.948629
Mean precision	0.980357
Mean recall	0.884090
India / US oracle F _{0.5}	0.94288 / 0.95244

Interpretation [PDD]. Oracle recall (0.884) exceeds pair recall (0.878) because macro-F_{0.5} is a per-entity mean: an entity that loses one of several links still scores high, so link-level recall loss is **sub-linear** in score loss. The **0.9486 ceiling is the number the end-to-end system is measured against** — the matcher's job (§16–17) is to convert as much of the 0.878→0.9486 headroom into realized score as precision allows.

Alternative policies bracket the trade [EXP]. The recall-leaning **75/75** policy raises the ceiling to **0.952327**; the compact **37/38** policy still holds **0.946593** at a much smaller candidate size. The frozen 50/50+heavy100 choice sits deliberately between them — near-max ceiling, tight size (O12). These alternatives are recorded for §32 (future policy v2), **not** grounds to unfreeze v1.

Four distinct evaluation layers [PDD] — never conflated:

Diagram source 4 (Mermaid — render via mermaid-cli for graphics)



L1/L2 are frozen and known today; a **pilot L3** exists (smoke matcher); **L4 requires the full matcher and is not computed**. Reporting L1 as if it were L4, or L4 as if it could exceed L2, is a category error this architecture forbids.

13. Matcher Split Architecture

Status: ✔ **Existing** (`matcher_split_checksums.json` , seed `matcher_split_v1_20260926`). A hard **leakage firewall** partitioning the labelled train S1 into four disjoint roles so tuning never contaminates evaluation (G7).

Split	S1 count	Role
<code>model_train</code>	1,765,608	fit the matcher
<code>model_calibration</code>	110,341	probability calibration + threshold τ tuning
<code>model_final_eval</code>	110,341	untouched held-out end-to-end macro- $F_{0.5}$
<code>candidate_dev</code>	220,531	candidate-policy development (the 0.878 recall split, §11)

Assignment mechanism [PDD]. Deterministic `MD5(entity_id || seed) → bucket` , so the split is reproducible from the id alone and stable across machines. Verified overlap: **cross-split S1 overlap = 0**; **labelled-target overlap = 0**.

Why S1-partitioning is leak-safe [verified fact, §7]. Because **each S2/S3 target matches ≤ 1 S1** (cross-S1 sharing = 0), partitioning on S1 also partitions the labelled targets — a target seen in `model_train` cannot reappear as a label in `model_final_eval` . This is the structural property that makes the firewall real, not nominal.

Honest scope of the firewall [per §36 — `foundation_gate_summary.md` correction]. Id-uniqueness prevents **target-record** leakage; it does **not** prevent **feature-distribution** leakage: **55,627** name-keys are shared across splits, and **46.31%** of val S1 share a name-key with train. Therefore the grouped-key holdout is a **stress-test diagnostic**, not a guaranteed lower bound on generalization. This correction is recorded so no future session mistakes "no target leakage" for "no leakage of any kind."

Distribution preserved [EXP]. Each split retains `mean_links` $\approx$ **3.46**, the singleton rate, and the US/India country mix (per-split stats in the checksums file), so calibration and eval see the true match-count distribution.

Diagram D4 — Split / leakage firewall:

```
flowchart TD
    GT["Train S1 + GT<br/>2,206,821 S1"] --> H["MD5(entity_id || seed)<br/>deterministic bucket"]
    H --> MT["model_train<br/>1,765,608"]
    H --> MC["model_calibration<br/>110,341 (τ tuning)"]
    H --> ME["model_final_eval<br/>110,341 (locked)"]
    H --> CD["candidate_dev<br/>220,531 (policy, §11)"]
    MT --> ME["S1 overlap 0<br/>target overlap 0"]
```

14. Feature Architecture

Status: 🟡 **In progress — pilot computed, full-scale materialization underway.** `feature_spec_v1` defines **65 label-free numeric features** in **7 groups**. The matrix is **no longer a scaffold**: a **pilot 65-d matrix has been computed** (`feature_pilot_audit.json` : **1,098,300 candidate rows**, added/removed/dup **0/0/0**, null **0**, nan/inf **0**, positives 21,186 / negatives 1,077,114) and persisted (`feature_pilot_v1_final/*.parquet` , split by country). **Full-scale feature materialization over the model-development splits is actively running** (`model_development_features/` : `model_fit` and `baseline_dev` base features complete; "extra" features complete for India, US in a `.partial.parquet` at last inspection). This is a **discrepancy correction** from the earlier draft, which stated "no feature matrix has been computed" — see §36. Full production matrix over all 34,568,979 candidates is still pending; **no test-set features exist** and `model_final_eval` remains firewalled (§13).

#	Group	What it captures	Source
1	provenance	which of the 4 passes fired (bitmask decomposition, evidence_score, rank)	§10 bitmask
2	exact	exact-key agreements (name_nosuffix, sorted_name, full address)	normalize keys
3	missing_length	address-missing flags, name/address lengths, length ratios	raw fields
4	token	shared-token counts, Jaccard, IDF-weighted overlap, coverage	token sets
5	numeric	shared digit sequences (street numbers, unit ids)	address digits
6	fuzzy	rapidfuzz ratios (token_sort, token_set, partial) on name/address	rapidfuzz
7	script	writing-system agreement (Latin/Devanagari), transliteration-bridge flags	unicode class

Design constraints [PDD].

- **Label-free features only** — every feature is a pure function of the two records + corpus DF stats; no feature peeks at GT (fair-play O22, and prevents trivial leakage).
- **Country as agreement, not identity [ORG O9]** — a boolean `country_match` , never a one-hot of {US, India}. This is what lets the model transfer zero-shot to France: France pairs still produce a `country_match` signal even though no France row was ever trained on.
- **Missing-aware [O8]** — `either_address_missing` is explicit; it is the feature most correlated with the recall cliff (0.729 vs 0.885, §11), so the matcher must see it rather than infer it.
- **Compact dtypes [ENG]** — float32 / small ints; the matrix is built per hash-partition and streamed, never fully resident (§19).

Diagram D5 — Feature extraction (pair → 65-d vector):

```

flowchart LR
    PAIR["Candidate pair<br/>(S1 record, target record)"] --> G1["provenance"]
    PAIR --> G2["exact"]
    PAIR --> G3["missing_length"]
    PAIR --> G4["token"]
    PAIR --> G5["numeric"]
    PAIR --> G6["fuzzy (rapidfuzz)"]
    PAIR --> G7["script / translit"]
    G1 --> V["65-d float32 vector<br/>+ train-only label"]
    G2 --> V
    G3 --> V
    G4 --> V
    G5 --> V
    G6 --> V
    G7 --> V

```

15. Negative Sampling Architecture

Status: 🟡 **Pilot sampler built and run; full-scale pending.** A negative sampler **exists and has been executed at pilot scale**, producing three checksummed policy artifacts (`negative_samples/`): `hard.parquet` (113,411 rows, 15,072 pos / 98,339 neg, SHA `b97d10a0...`), `random.parquet` (113,411 rows, SHA `8d3a65b5...`), and `hybrid_source_balanced.parquet` — **SELECTED** (113,367 rows, 15,072 pos / 98,295 neg, SHA `c34c0f65...`). Hard/random use 20 negatives/S1; the selected hybrid uses up to 10 hard-ranked negatives from **each** target source (S2, S3) to balance provenance. **All recovered positives are retained in every policy.** This corrects the earlier "no sampler code exists yet" draft (§36). Full-scale negative sampling over the complete `model_fit` split is pending.

Core principle [PDD]. Negatives are drawn only from the frozen candidate set — never from the full S2US3 cross-product. The matcher's job is to discriminate *within the candidates it will actually see at inference*, so its training distribution must be the candidate distribution, not an artificial one.

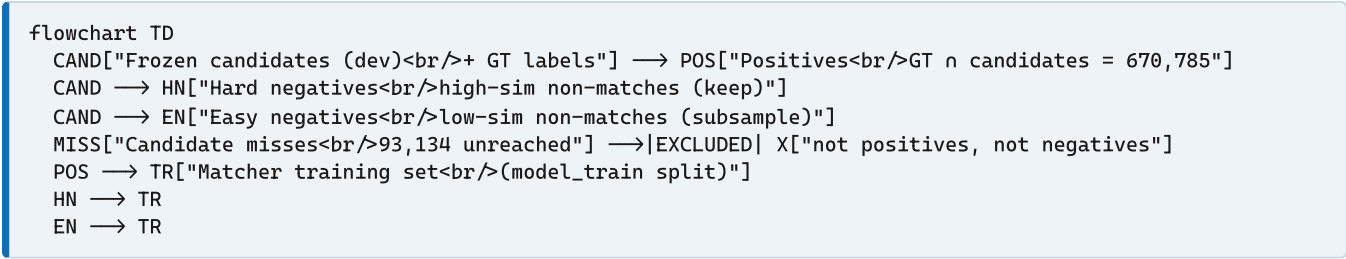
Label assignment on candidates [PDD].

- **Positives** = candidate pairs that are in the ground truth: `GT n candidates` . By construction these number **670,785** on the dev split (the recovered links, §11).
- **Hard negatives** = high-similarity candidate pairs that are **not** in GT (high fuzzy/token overlap, low `1/df` distinctiveness) — the pairs the matcher will most easily confuse; kept in full or over-weighted.
- **Easy negatives** = low-similarity candidate non-matches — **subsampling** to control class balance and matrix size (§19), since they are numerous and cheaply separable.

Explicit exclusion [PDD, critical]. Candidate-generation misses are NOT matcher negatives. A true link that blocking failed to retrieve (the ~93,134 unreachable, §11) is *absent from candidates*, so it can be neither a positive nor a negative — it is simply outside the matcher's universe. Treating misses as negatives would teach the model to reject true matches and is forbidden.

Class-balance policy [EXP-to-be]. Target ratio tuned on `model_calibration` (not `model_final_eval`); GBMs handle moderate imbalance with `scale_pos_weight` , so the sampler aims for a *bounded* negative multiple rather than 50/50. Final ratio is an experiment to log (§21), not a fixed assumption.

Diagram D6 — Negative sampling from candidates:



16. Matcher Architecture

Status: ● **In progress — pilot/smoke matchers fit; full-scale bake-off pending.** Contrary to the earlier " ■ Planned / no `.fit` " draft (corrected here per §36), **three matchers have been fit and thresholded at pilot scale** on the negative-sampled pilot set (113,367 sampled rows, 15,072 positives), scored on the calibration-dev pilot. These are **pilot/calibration-development numbers only — NOT leaderboard, NOT an end-to-end L4 score**, and `model_final_eval` was **never touched** (§13, §18):

Pilot matcher	τ	macro- $F_{0.5}$	Pair P	Pair R	Notes
Deterministic rule	0.56	0.628979	0.928970	0.426153	floor / feature sanity check
Logistic (linear logit)	0.93	0.781286	0.859481	0.702334	fixed-seed NumPy logit — Windows app-control blocked scikit-learn's compiled <code>_cd_fast</code> DLL
LightGBM smoke	0.67	0.851852	0.936502	0.734491	one fixed 40-tree / 15-leaf config, no search ; model → <code>work/lightgbm_smoke_v1.txt</code>

These artifacts (`baseline_results.json` , `logistic_baseline_v1.json` , `lightgbm_smoke.json`) confirm the progression below has been *piloted*, not that the production matcher is selected. **No hyperparameter search, no full-scale training, no calibration on the full split, and no test inference have run.**

Model choice [ORG O20 / PDD]. A **tabular gradient-boosted tree** (LightGBM 4.7.0 primary, XGBoost 3.4.1 comparator) over the 65-feature vector (§14). This satisfies the envelope **MIT/Apache-2.0, ≤8B params** trivially (a GBM is megabytes, not billions of params) and is **explicitly not an LLM** — the brief says an LLM is not the default, and a GBM is the right tool for dense low-dimensional tabular signal at hundreds-of-millions-of-pairs scale (O21).

Empirical progression [PDD — compare before choosing, per working rules].

1. **Deterministic rule baseline** — thresholded exact/fuzzy agreement; establishes a floor and sanity- checks features.
2. **Logistic regression** — calibrated linear baseline; cheap, interpretable, catches feature bugs.
3. **GBM (LightGBM / XGBoost)** — the intended production matcher; selected only if it beats the baselines on `model_calibration` by evidence, not assumption.

Output [PDD]. Per candidate pair, a **calibrated $P(\text{match}) \in [0,1]$** — **not** a forced top-1. This is mandatory: 76.8% of S1 match multiple records in one source (O3, data), so any `argmax/1:1` assignment structurally caps recall. Calibration (isotonic/Platt on `model_calibration`) makes the probability comparable across entities so a single global threshold τ (§17) is meaningful.

Scale execution [ENG O21]. Inference runs **per hash-partition** (16 shards): load one partition's feature matrix, score, write `P(match)` , release. Peak RSS stays bounded (§19); batched scoring, float32 throughout.

Model card at selection [ORG]. When the production model is chosen, a card records: family, license, param count, features, calibration method, `model_final_eval` macro- $F_{0.5}$, and the deterministic seed — the O20/O23 compliance record. A **pilot smoke model** exists (`work/lightgbm_smoke_v1.txt` , 40 trees); the **production card is not written yet** because no model has been selected and `model_final_eval` is firewalled.

Diagram D7 — Matcher (candidate → calibrated probability):

Diagram source 8 (Mermaid — render via mermaid-cli for graphics)

```
flowchart LR
    FM["65-d feature matrix<br/>(per hash-partition)"] --> M["Tabular GBM<br/>LightGBM / XGBoost □"]
    M --> RAW["raw score"]
    RAW --> CAL["Calibration<br/>isotonic/Platt on model_calibration"]
    CAL --> P["Calibrated P(match) per pair<br/>NOT top-1"]
    P --> DEC["→ S1 decision policy (§17)"]
```

17. Entity-Level Decision Architecture

Status: ■ **Planned.** Converts per-pair `P(match)` (§16) into the final per-S1 match set `M(e)` under the $F_{0.5}$ objective. This is where the precision-heavy metric is actually *won or lost*.

Threshold policy [PDD]. A single **global, calibrated threshold** τ , tuned on `model_calibration` to maximize macro- $F_{0.5}$, then **locked** before touching `model_final_eval`. Because $\beta=0.5$ weights precision $\approx 2\times$ recall, the optimum τ is **deliberately high** — the system prefers to drop a marginal candidate than risk a false merge.

Per-candidate keep/drop [ORG O2/O3]. For each S1, keep **every** candidate with `P(match)  $\geq \tau$` — zero, one, or many, spanning S2 and S3 freely. **No argmax, no per-source cap at decision time.** The one-to-many structure is honored by thresholding, not ranking.

Singleton path [ORG O4 — first-class]. If **no** candidate clears τ , emit an **empty** match set. A correct empty prediction scores a full **1.0**; a single false positive on a true singleton scores **0.0**. With 5.6% true singletons, the empty path is a direct, high-leverage contributor to macro- $F_{0.5}$ — it is engineered explicitly, not left to fall out of thresholding.

Multi-match refinement [PDD/FUT]. Optional post-threshold logic to be validated on `model_calibration` only: e.g., a light per-S1 consistency check when accepted candidates disagree strongly. Any such rule ships **only** if it lifts held-out macro- $F_{0.5}$; otherwise plain thresholding stands.

Hard invariant [ORG O13]. The decision stage may only **filter** candidates. `M(e)  $\subseteq$  C(e)` always; it can never emit an id absent from `candidate_pairs.tsv`. This guarantees the submission's matches- $\subseteq$ -candidates check passes by construction.

Diagram D8 — S1-level decision:

Diagram source 9 (Mermaid — render via mermaid-cli for graphics)

```
flowchart TD
    P["Calibrated P(match)<br/>for each candidate of S1 e"] --> T["P  $\geq \tau$  ?<br/>(τ tuned on calibration)"]
    T --> S["none clear τ | S[\"Emit empty set<br/>singleton = 1.0 if true (O4)\"]"]
    T --> K["≥1 clears τ | K[\"Keep all that clear τ<br/>0..many, S2+S3 (O2/O3)\"]"]
    K --> INV["Invariant: M(e)  $\subseteq$  C(e) (O13)"]
    S --> OUT["matching_results.tsv row"]
    INV --> OUT
```

18. Evaluation Architecture

Status:  **Existing, FROZEN.** `code/.../evaluate.py`, SHA-256 `e99d7f0a...` (pinned frozen input). It implements macro- $F_{0.5}$ **exactly** per O17/O18 and **reproduces the spec worked example** → **0.714**, which is the correctness gate for the whole scoring stack.

What it computes [ORG O17/O18]. Per S1: $P = |\text{predntruth}|/|\text{pred}|$, $R = |\text{predntruth}|/|\text{truth}|$, $F_{0.5} = (1.25 \cdot P \cdot R) / (0.25 \cdot P + R)$; then the **macro-average** over all S1. Edge cases hard-coded: empty truth $\wedge$ empty pred = **1.0**; non-empty pred on empty truth = **0.0**; empty pred on non-empty truth = 0.0.

Four evaluation layers, four purposes [PDD — §12]:

Layer	Question	Number today	Uses labels?
L1 candidate pair recall	did blocking retrieve the true links?	0.878084	dev GT (diagnostic, O19)
L2 candidate oracle $F_{0.5}$	best possible score on this candidate set?	0.948629	dev GT (ceiling)
L3 matcher pair P/R	is the classifier accurate per pair?	 pilot (GBM-smoke P 0.9365 / R 0.7345)	pilot calibration-dev GT
L4 end-to-end macro- $F_{0.5}$	the actual leaderboard-shaped score		<code>model_final_eval</code> (locked)

Pilot L3 exists; L4 does not [per §36]. The smoke matcher produced **pilot** pair P/R and per-entity macro- $F_{0.5}$ (0.851852) on the **calibration-development** pilot — an L3-type diagnostic, explicitly **not** L4. `model_final_eval` was never scored.

Discipline [ORG O19 / PDD]. L1 is a **diagnostic, not the score** — recall $\neq$ leaderboard. Only **L4 on `model_final_eval`** is the honest estimate of competition performance, and it is reported **once**, after τ is locked, to avoid overfitting the eval split (§13). Only a **pilot L3** diagnostic exists; **no L4 number exists yet and none is claimed** (§36).

Country-wise + open-set reporting [ORG O9]. Evaluation is sliced by country. For the **France** zero-shot path, planned **leave-one-country-out (LOCO)** probes train on {US} or {India} and evaluate on the held-out country as a proxy for France generalization (, §25). Today only candidate-layer country recall is known (India 0.8758 / US 0.8796, §11).

Release gate [ORG O16]. Separately from scoring, `validate_submission.py` (§26) is the format gate; a run is release-eligible only when it prints **PASS** and the L4 estimate is recorded.

19. Resource and Scalability Architecture

Dominant constraint [ENG]. ~0.8 GB free RAM against ~5M-row string tables and **hundreds of millions** of candidate pairs pre-prune (O21). Every stage is designed out-of-core.

Out-of-core techniques [ENG].

- **DuckDB** for big joins with a bounded `memory_limit` + `temp_directory` → hash joins/sorts **spill to disk** rather than OOM.
- **polars streaming/lazy** scans; **pyarrow** Parquet as the on-disk columnar format.
- **Integer-coded IDs, float32 / small-int** feature dtypes, **scipy sparse** where applicable.

- **16 hash partitions** (`hash(source1_entity_id) % 16`) so no stage materializes the full 34.6M-row candidate table; features and inference run **per shard**.
- **Atomic .partial writes + per-part SHA-256** → restart/resume without corruption (§20).

Measured memory — reported honestly, NOT conflated [per §36]. Distinct operations have distinct peaks; none should be described as "comfortable":

Operation	Peak RSS	Note
Production candidate reproduction	≈ 648.5 MiB	temp spill 0, exit 0, identity 0/0/0 (§11)
Foundation steps	600–680 MiB	ingest/normalize/split
Full candidate_generation audit (full materialization)	≈ 1.27 GiB	DuckDB spill enabled
Projected full-scale test inference	< 1.1 GiB	projection,  not yet measured

Discrepancy surfaced [per §36]. The audit peak (≈**1.27 GiB**) and projected inference (<**1.1 GiB**) **exceed the ~0.8 GB planning budget**. They complete only because DuckDB spills to disk; the budget is a *resident* target, not a hard ceiling on peak. This is a real risk for full-scale test inference (§27) and must be validated, not assumed away.

A prior "137 MB" claim was corrected [per §36 — foundation_gate_summary.md] to the true process peak ≈**600–680 MiB**; the smaller figure measured only a sub-step, not the process.

Diagram D9 — Resource / partition strategy:

Diagram source 10 (Mermaid — render via mermaid-cli for graphics)

```
graph TD
    TD["flowchart TD"]
    TD --> BIG["BIG[\"S1x(S2uS3) evidence<br/>hundreds of millions pre-prune\"] → DUCK[\"DuckDB bounded memory_limit<br/>+ temp_directory spill\"]"]
    TD --> DUCK["DUCK → HP[\"16 hash partitions<br/>hash(source1_entity_id)%16\"]"]
    TD --> HP["HP → S0[\"shard 0\"]"]
    TD --> HP["HP → SE[\"...\"]"]
    TD --> HP["HP → S15[\"shard 15\"]"]
    TD --> S0["S0 → PROC[\"per-shard: features → score<br/>load, compute, write, release\"]"]
    TD --> S15["S15 → PROC"]
    TD --> PROC["PROC → ATOM[\".partial atomic write<br/>+ per-part SHA-256 (resume-safe)\"]"]
```

20. Reproducibility Architecture

Status:  **Existing for frozen stages.** Reproducibility is enforced by **content-addressed manifests**, fixed seeds, and pinned versions — the O23 compliance mechanism.

Manifest chain [PDD]. Every frozen artifact pins the SHA-256 of its inputs *and* its outputs, so the whole candidate stage is verifiable end-to-end:

- `final_candidate_policy_manifest.json` — pins **normalize.py** `b3508b60...`, **evaluate.py** `e99d7f0a...`, `split_s1.parquet` `d01e4811...` (2,206,821 rows), `evidence_ranker_v1.json` `a0fb606f...`, `rank_shards_e2db9fd4...` (106,487,165 rows / 128 parts); policy `46fd324b...` .
- `final_candidate_artifact_manifest.json` — 16 parts (part-00...15) with per-part rows/bytes/sha256; row_count **34,568,979**; partition_manifest_sha256 `221fef24...9465b067` .
- `matcher_split_checksums.json` — split seed `matcher_split_v1_20260926` , per-split counts + `entity_id_sha256` .

Freeze gate [PDD]. Regenerating the candidate artifact and diffing identities yields **0 added / 0 removed / 0 duplicate**, `equal=true` → freeze gate **passed**. This is the objective proof the artifact is reproducible bit-for-bit from `code/` , and the reason it may be treated as an immutable contract (§11).

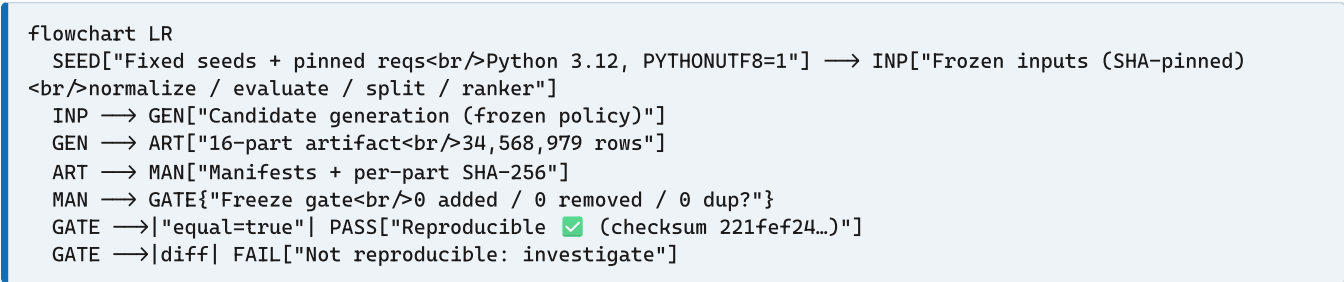
Determinism controls [PDD/ENG].

- **Fixed seeds** — candidate-dev split seed 42; matcher split seed `matcher_split_v1_20260926` ; model seeds to be pinned at training ().
- **Pinned versions** — `requirements.txt` pins numpy 2.5.3 / pandas 3.0.6 / scikit-learn 1.9.1 / rapidfuzz 3.14.6 / lightgbm 4.7.0 / xgboost 3.4.1 / polars 1.44.2 / pyarrow 25.0.1 / duckdb 1.5.5.
- **Deterministic hashing** — `MD5(entity_id||seed)` splits and `hash()%16` partitions reproduce identically.
- **Runtime** — `.venv` Python **3.12** (system 3.14 lacks wheels); `PYTHONUTF8=1` ; `sys.stdout.reconfigure(utf-8, replace)` in every script (Windows UTF-8 safety).

Reproduction command [ORG O23]. The policy manifest carries the exact regeneration command; running it must reproduce the checksum `221fef24...9465b067` . Experiment tracking (§21) is append-only, so any run is auditable after the fact.

Diagram D10 — Reproducibility / artifact chain:

Diagram source 11 (Mermaid — render via mermaid-cli for graphics)



21. Experiment Tracking

Status:  **Existing (append-only ledger)** +  **narrative reports.** Every measured claim in this document traces to a logged run, per the working rule *measure, never assume*.

Ledger [PDD]. `work/ledger.duckdb` — an **append-only** DuckDB store of runs: policy/experiment id, inputs (with SHA), metrics (recall, size, reduction-ratio, oracle $F_{0.5}$, runtime, peak RSS), and outcome. Append-only so history cannot be silently rewritten to fit a narrative.

Narrative reports [PDD] — the human-readable evidence trail:

Report	What it fixes as evidence
<code>integrity_report.md</code>	dataset scale, GT distribution, referential integrity (§7)
<code>baseline_blocking.md</code>	exact-block recall 0.4702 / sorted 0.5061 (§9 justification)
<code>candidate_experiment_summary.md</code>	single-pass + cumulative recall; intermediate 0.8614 (§10)
<code>candidate_refinement_report.md</code>	13 policies; selected 0.878084 / 34.6M (§11)
<code>candidate_oracle_report.md</code>	ceiling 0.948629 + alternatives (§12)
<code>foundation_gate_summary.md</code>	corrections: memory ~600–680 MiB; leakage scope (§13, §19)

Why not MLflow [PDD]. A hosted tracker adds a service dependency and RAM overhead for no benefit at this scale; a checksummed append-only DuckDB ledger + committed markdown reports give full auditability within the memory budget and the "runnable from `code/` " constraint (O23). Recorded as a deliberate decision (ADR-011, §29), revisitable if the team grows.

Discipline encoded [PDD]. A claim enters this architecture only with a citable run; corrections (the two in `foundation_gate_summary.md`) are logged as first-class events, not edits — so "137 MB→ 600–680 MiB" and the leakage-scope re-scoping are part of the permanent record (§36).

22. Repository Architecture

Principle [ORG O23]. Everything needed to reproduce the submission lives under `code/`; `work/` holds manifests/reports/ledger; `dataset/` and `output/` are data in/out. Status markers reflect **actual** repo evidence (§31), not intent.

Path	Role	Status
<code>code/ ... /normalize.py</code>	frozen deterministic normalization (SHA <code>b3508b60...</code>)	✓
<code>code/ ... /evaluate.py</code>	frozen macro-F _{0.5} evaluator (SHA <code>e99d7f0a...</code> , →0.714)	✓
<code>code/ ... /split.py</code>	candidate-dev split (seed 42) + matcher split v1	✓
<code>code/ ... /materialize.py</code>	16-partition candidate materialization (out-of-core)	✓
<code>matcher/</code> split tables + firewall; pilot matchers (<code>lightgbm_smoke_v1.txt</code> , logistic, deterministic) fit	pilot done; full-scale 	
feature extraction	<code>feature_spec_v1</code> 65-d; pilot matrix computed (1,098,300 rows, audit 0/0/0); full-scale materialization running	
negative sampler	candidate-only sampling; 3 policies run (<code>negative_samples/*.parquet</code>), <code>hybrid_source_balanced</code> selected	
S1 decision policy	threshold + singleton + multi-match	
submission assembler	emits the two TSVs, exact format	
<code>utils/validate_submission.py</code>	official format gate (→PASS)	✓
<code>scripts/integrity_check.py</code> , <code>scripts/profile_dataset.py</code>	profiling/integrity	✓
<code>work/*.json</code> , <code>work/*.md</code> , <code>work/ledger.duckdb</code>	manifests, reports, experiment ledger	✓
<code>dataset/{train,test}/</code>	inputs (S1/S2/S3 [+GT train])	✓ (data)
<code>output/</code>	<code>matching_results.tsv</code> , <code>candidate_pairs.tsv</code>	 empty

CLI surface [PDD]. A single `er` entrypoint with subcommands mirrors the pipeline stages: `split` | `analyze-gt` | `block` | `features` | `train` | `predict` | `evaluate` | `submit` | `validate`. Stages `split` / `block` are backed by existing code; `features` / `train` / `predict` / `submit` are the planned modules above.

Module boundaries [PDD]. The three hard concerns — **candidate generation** (frozen), **pairwise scoring**, **entity decision** — are separate modules with separate manifests and separate evaluation layers (§12/§18), so freezing one does not entangle the others. This is what lets policy v1 be immutable while the matcher is still unbuilt.

23. Technology Stack

All libraries are permissive-licensed (MIT/BSD/Apache), satisfying O20 for the shipped stack. Versions are pinned (§20). Status = whether the component is *actively used today* (✅) or *reserved for a planned stage* (🟡).

Layer	Technology	Role	Why Used	Status
Runtime	Python 3.12 (<code>.venv</code>)	interpreter	system 3.14 lacks wheels; 3.12 has all binaries	✓
Normalization	stdlib <code>unicodedata</code>	deterministic folding/translit	fair-play: transform, never look up (O22)	✓
Columnar store	pyarrow 25.0.1 / Parquet	typed on-disk store	avoids loading 5M-row string tables into RAM	✓
Out-of-core SQL	duckdb 1.5.5	big joins, spill to disk	joins hundreds of M pairs under 0.8 GB budget (O21)	✓
DataFrame	polars 1.44.2	streaming/lazy scans	low-memory columnar ops	✓
Fuzzy strings	rapidfuzz 3.14.6	token_sort/set/partial ratios	fast C++ fuzzy features (§14 fuzzy group)	● (contract)
Numerics	numpy 2.5.3 / scipy 1.18.1	arrays, sparse matrices	compact dtypes, sparse overlap	✓ / ●
GBM (primary)	lightgbm 4.7.0	tabular matcher	≤8B, fast, MIT; not an LLM (O20)	● smoke fit (<code>lightgbm_smoke_v1.txt</code>)
GBM (comparator)	xgboost 3.4.1	matcher comparison	empirical model bake-off	■
Calibration/LR	scikit-learn 1.9.1	logistic baseline, isotonic/Platt	calibrated P(match), baselines (§16)	⚠ compiled <code>_cd_fast</code> DLL blocked by Windows app-control; pilot LR ran via fixed-seed NumPy logit
Experiment log	duckdb (<code>work/ledger.duckdb</code>)	append-only run ledger	auditable, no service dependency (§21)	✓
Validation	<code>utils/validate_submission.py</code>	official format gate	O16 PASS gate	✓
Docs/diagrams	mermaid-cli (mmdc) 12.0.0 , node 24.19.0	render diagrams → PNG, md → PDF	local, no external service (§26 build)	✓ (tooling)

Deliberately excluded [PDD]. No deep-learning framework, no vector DB, no hosted tracker, **no LLM** in the scored pipeline — each would add memory/latency/licensing/fair-play risk without evidence of benefit at this scale. `pandas 3.0.6` is available but **not** used for full-table string loads (ENG rule); DuckDB/polars handle scale.

24. End-to-End Implementation Plan

Phases with status from **actual repo evidence** (§31); COMPLETED requires a verified artifact. Traceable to `ULTRA_PLAN.md` (Phases 0–18) and `IMPLEMENTATION_PLAN.md` (ER-001...ER-032, a backlog with no done-markers).

Phase	Work	Status	Evidence
0	Environment, pinned deps, UTF-8 harness	COMPLETED	<code>.venv 3.12</code> , <code>requirements.txt</code>
1	Ingest + profile + integrity	COMPLETED	<code>integrity_report.md</code> (0 violations)
2	Frozen normalization	COMPLETED	<code>normalize.py</code> SHA <code>b3508b60...</code>
3	Frozen evaluator (→0.714)	COMPLETED	<code>evaluate.py</code> SHA <code>e99d7f0a...</code>
4	Baseline blocking (recall 0.47)	COMPLETED	<code>baseline_blocking.md</code>
5	Four-pass retrieval + IDF rank	COMPLETED	<code>candidate_experiment_summary.md</code>
6	Policy refinement + freeze v1	COMPLETED	<code>candidate_refinement_report.md</code> , freeze gate
7	Candidate oracle ceiling	COMPLETED	<code>candidate_oracle_report.md</code> (0.9486)
7b	Matcher split + firewall	COMPLETED	<code>matcher_split_checksums.json</code> (overlap 0)
8	Feature matrix computation (65-d)	PILOT DONE / full-scale IN PROGRESS	pilot 1,098,300 rows (audit 0/0/0); <code>model_development_features/</code> materializing 🟡
9	Negative sampling from candidates	PILOT DONE	3 policies run; <code>hybrid_source_balanced</code> selected 🟡
10	Matcher bake-off (rule→LR→GBM)	PILOT/SMOKE DONE / full-scale NEXT	pilot fits 0.629/0.781/0.852; no search yet 🟡
11	Calibration + threshold τ tuning	NEXT	on <code>baseline_dev / model_tune / model_threshold</code> 🟡
12	S1 decision policy (singleton/multi)	NEXT	🟡
13	LOCO / France zero-shot probes	FUTURE	\$25 🟡
14	Full-scale test inference (per shard)	FUTURE	memory risk \$27 🟡
15	Submission assembly (two TSVs)	FUTURE	<code>output/</code> empty 🟡
16	Validate → PASS + final L4 report	FUTURE	gate \$26 🟡

Critical path now [PDD]. Phase 8 (feature computation) unblocks 9→10→11→12; only after a τ is locked on `model_calibration` do Phases 13–16 run. **No phase ≥8 may touch `model_final_eval`** except the single final L4 read (§13/§18).

Diagram D11 — Training pipeline (build-time):

Diagram source 12 (Mermaid — render via mermaid-cli for graphics)

```
flowchart TD
  A["Frozen candidates + labels<br/>(model_train split)"] --> B["Phase 8 features 65-d 🟡"]
  B --> C["Phase 9 negatives from candidates ☐"]
  C --> D["Phase 10 GBM bake-off ☐"]
  D --> E["Phase 11 calibrate + tune τ<br/>on model_calibration ☐"]
  E --> F["Phase 12 decision policy ☐"]
  F --> G["Lock model + τ → single L4 read<br/>on model_final_eval (§18)"]
```

25. Final Production Pipeline

Status: ■ **Planned** (frozen stages ✓ feed it). This is the **test-time** path that produces the two submission files for the **1,732,544** test S1 (incl. **259,452** France).

Steps [PDD].

1. **Ingest + normalize test** — same frozen `normalize.py` (✓) on test S1/S2/S3; identical keys.
2. **Candidate generation on test** — apply the **frozen policy v1** (§11) to produce test candidates, 16 hash partitions. `candidate_pairs.tsv` is written **directly from this exact set** (O11/O13).
3. **Feature extraction** — 65-d vectors per candidate, per shard (§14, Phase 8).
4. **Matcher scoring** — locked GBM → calibrated `P(match)` per candidate, per shard (§16).
5. **Decision** — global locked τ ; keep-all- $\geq \tau$; empty-set singleton path (§17).
6. **Assemble** — emit `matching_results.tsv` ($M(e) \subseteq C(e)$) and `candidate_pairs.tsv`; exact format (§26).
7. **Validate** — `validate_submission.py` must print **PASS** (§26, O16).

France zero-shot handling [ORG O9 / PDD]. No code path special-cases France. It generalizes because: (a) normalization is country-agnostic (accent-fold covers France, §9); (b) candidate generation is label-free and key-based, so France blocks like any country; (c) the matcher sees **country_match as agreement**, never a {US, India} one-hot, so a France pair is in-distribution for the features even though no France row was trained on. **LOCO probes** (Phase 13) estimate the generalization gap before trusting the France slice.

Inference resource note [ENG/§27]. Step 4 is the memory-risk stage (projected <1.1 GiB vs 0.8 GiB budget, §19); it runs strictly **per shard** with release-between-shards and DuckDB spill, and must be **measured** at full scale before the final run — not assumed.

Diagram D12 — Test / inference pipeline (produces both TSVs):

Diagram source 13 (Mermaid — render via mermaid-cli for graphics)

```
flowchart TD
  T["Test S1/S2/S3<br/>1,732,544 S1 (France 259,452)"] --> NZ["Normalize (frozen) ✓"]
  NZ --> CGT["Candidate gen - frozen policy v1 ✓<br/>16 shards"]
  CGT --> CPQ["candidate_pairs.tsv (exact set) O11/O13"]
  CGT --> FET["Features 65-d per shard 🟡"]
  FET --> SC["GBM → calibrated P(match) ☐"]
  SC --> DEC["Global τ + singleton path ☐"]
  DEC --> MRT["matching_results.tsv"]
  MRT --> V["validate_submission.py → PASS ✓"]
  CPQ --> V
```

26. Submission Architecture

Status: assembler  / validator . Two files, both graded (O10–O13), both format-gated (O14–O16).

Exact output contract [ORG].

- `output/matching_results.tsv` — header `source1_entity_id→matched_entity_ids` (real TAB); **one row per test S1**; empty field = singleton.
- `output/candidate_pairs.tsv` — header `source1_entity_id→candidate_entity_ids`; the **exact** frozen candidate set the matcher scored (§11/§25).

Writer rules the assembler must obey [ORG O14/O15] (any breach = rejection):

- Real TAB separator; comma-joined lists with **no space after commas**.
- **No leading/trailing whitespace** on the S1 key or ids (validator does not strip).
- Only S2-/S3- ids; no S1 self-match; no duplicate id in a list; no duplicate S1 row; **every** test S1 present; no id absent from the test set.
- $M(e) \subseteq C(e)$ for every row (O13) — guaranteed because decision only filters candidates (§17).

Release gate [ORG O16].

```
.venv/Scripts/python utils/validate_submission.py --matching output/matching_results.tsv --candidate ou
```

Must print **PASS**. A run is release-eligible only when the validator passes **and** the L4 `model_final_eval` macro-F_{0.5} is recorded (§18).

Diagram D13 — Submission / validation:

Diagram source 14 (Mermaid — render via mermaid-cli for graphics)

```
flowchart TD
    DEC["Decision output M(e)"] --> WM["Write matching_results.tsv<br/>exact format, no strip"]
    FROZ["Frozen candidate artifact"] --> WC["Write candidate_pairs.tsv<br/>exact set"]
    WM --> SUB{"validate_submission.py"}
    WC --> SUB
    SUB -->|PASS| REL["Release-eligible<br/>( + record L4 macro-F0.5 )"]
    SUB -->|FAIL| FIX["Reject: fix format, re-run"]
```

27. Failure Modes and Safeguards

Failure Mode	Detection	Prevention	Recovery
OOM at full-scale inference (peak >0.8 GB budget; §19)	RSS monitor per shard; DuckDB spill counter	16-shard per-partition processing; bounded <code>memory_limit + temp_directory</code> ; float32	resume from last <code>.partial</code> shard; shrink shard/batch; add spill
False merge ($\beta=0.5$ costs $\approx 2\times$)	precision on <code>model_calibration</code> ; per-S1 $F_{0.5}$	high τ ; precision-weighted tuning; hard-negative training (§15)	raise τ ; re-tune on calibration only
Over-predicting on true singletons ($\rightarrow 0.0$)	singleton-slice $F_{0.5}$	first-class empty path; conservative τ (§17)	raise τ ; audit singleton false-positives
Candidate recall ceiling (12.2% links unreachable; §11)	oracle vs recall layers (§12)	accepted, bounded trade; not fixable without unfreezing v1	policy v2 experiment (§32) — never edit v1
France zero-shot collapse (0 train rows; O9)	LOCO probes (Phase 13)	country-agnostic normalize; agreement-only country feature	fall back to deterministic-rule matcher for France slice
Split leakage (feature-distribution, not target; §13)	cross-split name-key audit (55,627 shared)	S1-partition (target-safe); grouped-key stress test	report as diagnostic; discount optimistic val
Submission format rejection (O14/O15)	<code>validate_submission.py</code> → FAIL	exact-format writer; no strip; subset check	fix writer; re-validate before release
Matches $\nsubseteq$ candidates (O13 breach)	validator subset check	decision only filters candidates (§17 invariant)	drop offending ids; re-emit
Non-reproducible artifact (O23)	freeze gate identity diff	SHA-pinned manifests; fixed seeds; pinned reqs	regenerate from <code>code/</code> ; diff to <code>221fef24...</code>
Fair-play violation (external lookup; O22)	code review; <code>label_use=false</code> assertion	stdlib-only normalize; no network in pipeline	remove offending path; disqualifying if shipped
Eval-split overfitting	single locked L4 read	τ tuned only on calibration; <code>model_final_eval</code> locked	never re-tune against final eval

Standing safeguard [PDD/§28]. The frozen stages (normalize, evaluate, candidate policy v1, splits) are checksum-guarded; any accidental edit is caught by a manifest mismatch **before** it can propagate into a submission.

28. Security / Data Governance Constraints

Fair-play is the top governance rule [ORG O22 — disqualifying]. Learning comes **only** from the provided data. Prohibited and absent from every code path: external business-identity lookup, enrichment, commercial ER APIs, government/business registries, geocoding APIs, external company datasets, any internet-based entity enrichment. **Permitted:** deterministic local normalization via `stdlib unicodedata` . The distinction is enforced conceptually and asserted in the policy manifest (`label_use = all false`); transforming a string is allowed, **resolving a real business is not**.

No network in the scored pipeline [PDD]. The candidate→feature→matcher→decision→submit path makes no outbound calls. Web access is reserved for **generic technical documentation only**, never to resolve/enrich a business.

Label governance [PDD/§13]. Ground truth exists train-only. The matcher-split firewall ensures GT touches `model_train / model_calibration` for fitting/tuning and reaches `model_final_eval` **once** for the honest estimate. **Test labels are never accessed** (there are none to access) and **no full-scale test inference has been run** (§31/§37).

Immutability governance [PDD/§20]. Frozen inputs (normalize `b3508b60...`, evaluate `e99d7f0a...`, split `d01e4811...`, ranker `a0fb606f...`) and the candidate artifact (`221fef24...9465b067`) are SHA-pinned. Editing any of them **must** be caught by a manifest mismatch; freezing exists precisely so a later session cannot silently alter the contract (§36).

Secrets / PII [ENG]. The data is business names/addresses, not credentials; no `.env` /secret files are read by the pipeline. Business text is handled as data, emitted only into the two output TSVs.

Model licensing [ORG O20]. Only MIT/Apache-2.0, ≤8B components ship. The tabular GBM plan and every listed library (§23) comply; an LLM is excluded from the scored pipeline.

Reproducibility-as-governance [ORG O23]. Because the whole frozen stage is content-addressed and the ledger is append-only, any reviewer can independently verify that what ran is what is documented — the integrity guarantee behind every number in this document.

29. Architecture Decision Records

Each ADR: **Context / Decision / Alternatives / Consequences / Evidence-status.**

ADR-001 — Retrieve-then-score with a hard frozen boundary.

- *Context:* one-to-many ER over 2M+×10M+ records under 0.8 GB RAM; blocking is separately graded (O12).
- *Decision:* three separated concerns — frozen candidate generation → pairwise matcher → S1 decision — each with its own manifest and evaluation layer.
- *Alternatives:* monolithic end-to-end learned matcher; single-stage rules.
- *Consequences:* candidate stage can be frozen/immutable while the matcher is still unbuilt; clean freeze gates; slight duplication of key logic across stages.
- *Evidence:* **Strong** — freeze gate passed, four eval layers operational (§11/§12).

ADR-002 — Freeze candidate policy v1 as an immutable upstream contract.

- *Context:* candidate set is a graded deliverable and the recall ceiling for everything downstream.
- *Decision:* freeze `evidence_source_50_50_heavy100_v1` (34,568,979 rows, recall 0.878084) and forbid edits; downstream builds against its checksum.
- *Alternatives:* keep candidates fluid until the matcher exists; co-optimize blocking+matching jointly.
- *Consequences:* stable target for feature/matcher work; reproducible; **caps recall at 0.878** (bounded loss, §6/§11); improvements require a *new* policy v2, never an edit.
- *Evidence:* **Strong** — identity reproduction 0/0/0, `equal=true`.

ADR-003 — Optimize macro-F_{0.5} directly, precision-first.

- *Context:* $\beta=0.5 \Rightarrow$ precision $\approx 2 \times$ recall; correct singleton = 1.0; false merge = 0.0.
- *Decision:* high locked τ , precision-weighted tuning, first-class singleton path.
- *Alternatives:* F₁-style balance; recall-max blocking then aggressive keep.

- *Consequences*: conservative predictions; singleton accuracy directly lifts score; some recall deliberately foregone.
- *Evidence*: **Strong (metric)** — `evaluate.py` reproduces spec 0.714; τ tuning .

ADR-004 — Country as an agreement feature, never a one-hot.

- *Context*: test adds France with **0** training rows (open set, O9).
- *Decision*: single `country_match` boolean; no {US, India} encoding anywhere.
- *Alternatives*: one-hot country; per-country models.
- *Consequences*: France pairs stay in-distribution zero-shot; slightly less country-specific signal.
- *Evidence*: **Medium** — design enforced; LOCO/France probes  (\$25).

ADR-005 — Tabular GBM as the matcher, not an LLM.

- *Context*: O20 caps the model at MIT/Apache-2.0, $\leq 8B$, LLM-not-default; signal is dense low-dim tabular.
- *Decision*: LightGBM primary, XGBoost comparator, over 65 features; LR + rule baselines first.
- *Alternatives*: fine-tuned small LLM; deep tabular nets; pure rules.
- *Consequences*: megabyte-scale model, fast per-shard inference under budget, trivially compliant; needs good hand-designed features.
- *Evidence*: **Medium** — envelope fit certain; a **pilot LightGBM-smoke fit exists** (macro- $F_{0.5}$ 0.851852 on the calibration-dev pilot, `lightgbm_smoke.json`), but the production bake-off/search is not run (\$16, \$36).

ADR-006 — Per-candidate thresholding, never top-1.

- *Context*: mean 3.46 matches/S1; 76.8% match multiple within one source (O3).
- *Decision*: keep every candidate with $P(\text{match}) \geq \tau$; 0..many; no argmax/1:1.
- *Alternatives*: top-1; per-source top-k.
- *Consequences*: honors one-to-many; requires calibrated cross-entity probabilities.
- *Evidence*: **Strong (structural)** — GT distribution measured (\$7); mechanism .

ADR-007 — Negatives drawn only from the candidate set.

- *Context*: matcher only ever sees candidates at inference; misses are outside its universe.
- *Decision*: positives = $GT \cap \text{candidates}$; hard negatives = high-sim non-matches; easy negatives subsampled; **candidate misses excluded**.
- *Alternatives*: sample negatives from full cross-product; treat misses as negatives.
- *Consequences*: train distribution = inference distribution; avoids teaching rejection of true matches.
- *Evidence*: **Medium** — design fixed; sampler  (\$15).

ADR-008 — Leakage firewall by deterministic S1 partition.

- *Context*: tuning must not contaminate the honest estimate; each S2/S3 target maps to ≤ 1 S1.
- *Decision*: `MD5(entity_id||seed)` $\rightarrow$ {model_train, model_calibration, model_final_eval, candidate_dev}; `model_final_eval` read once.
- *Alternatives*: random row split; k-fold on pairs.
- *Consequences*: target-leak-free by construction; **does not** remove feature-distribution overlap (55,627 shared name-keys) — grouped-key holdout is a stress test, not a bound.
- *Evidence*: **Strong** — overlaps 0/0 verified; leakage-scope correction logged (\$13).

ADR-009 — Out-of-core everything; 16 hash partitions + DuckDB spill.

- *Context*: ~0.8 GB free RAM vs hundreds of millions of pairs (O21/ENG).
- *Decision*: Parquet columnar, integer IDs, float32, DuckDB `memory_limit` + spill, 16 shards, atomic `.partial` + per-part SHA.
- *Alternatives*: in-RAM pandas; a bigger machine (unavailable).

- *Consequences*: fits budget with disk spill; resume-safe; **peaks still exceed 0.8 GB** on full materialization (≈ 1.27 GiB) and projected inference (< 1.1 GiB) — must be measured (§19/§27).
- *Evidence*: **Strong (reproduction ≈ 648.5 MiB, spill 0) + flagged risk** at full scale.

ADR-010 — Deterministic stdlib-only normalization, frozen.

- *Context*: noise (translit, accents, suffixes, junk) must collapse without any external lookup (O22).
- *Decision*: `unicodedata`-based fold/translit/suffix-canon in `normalize.py`, frozen at `b3508b60...`.
- *Alternatives*: ML transliteration; external name-canonicalization services (**disqualifying**).
- *Consequences*: fair-play-safe, reproducible, lifts recall $0.47 \rightarrow 0.88$; frozen so candidate identity holds.
- *Evidence*: **Strong** — baseline vs four-pass recall measured (§9/§10).

ADR-011 — Append-only DuckDB ledger + markdown reports, not MLflow.

- *Context*: need auditable experiment history within the memory budget and "runnable from `code/`".
- *Decision*: `work/ledger.duckdb` (append-only) + committed narrative reports; no hosted tracker.
- *Alternatives*: MLflow / W&B (service + RAM overhead).
- *Consequences*: zero service dependency, full auditability, corrections logged as events (§21); no fancy UI.
- *Evidence*: **Strong** — reports back every number in this document.

ADR summary map.

ADR	Decision	Evidence
001	retrieve-then-score, frozen boundary	Strong
002	freeze candidate policy v1	Strong
003	precision-first macro- $F_{0.5}$	Strong (metric) / τ 
004	country as agreement only	Medium
005	tabular GBM, no LLM	Medium
006	per-candidate threshold, no top-1	Strong (structural)
007	negatives from candidates only	Medium
008	S1-partition leakage firewall	Strong
009	out-of-core, 16 shards + spill	Strong + risk flagged
010	frozen stdlib normalization	Strong
011	DuckDB ledger, not MLflow	Strong

30. Requirement Traceability Matrix

Full O1–O23 → component → implementation → validation → status. Status reflects §31 evidence.

#	Organizer Requirement	Source	System Component	Implementation	Validation	Status
O1	S1 = deduplicated reference	PS	pipeline key, assembler	key on S1; 1 row/S1	integrity_report (1 GT/S1)	✓
O2	0..many matches per S1	PS	decision policy	keep-all- $\geq \tau$	oracle recovered counts	■
O3	matches span S2 & S3	PS	decision policy	no per-source cap	GT 80.5% both	■
O4	singletons exist	PS	empty path	empty set if none $\geq \tau$	evaluator empty/empty=1.0	■
O5	record schema	PS/validator	<code>io.py</code>	typed columns	integrity_check	✓
O6	id prefixes, disjoint ns	PS	id handling	S1/S2/S3, disjoint	integrity (disjoint 0)	✓
O7	GT schema (train)	PS	GT loader	comma-join, empty=singleton	0 dup GT rows	✓
O8	noise classes	PS	<code>normalize.py</code> (frozen)	fold/translit/suffix	recall 0.47→0.88	✓
O9	open-set France	PS	country feature; LOCO	agreement only	country_open_set:true; probes ■	●
O10	matching_results.tsv	validator	assembler	exact header/TAB	validate → PASS	■
O11	candidate_pairs.tsv = exact set	validator	frozen artifact + writer	write from artifact	validate subset	●
O12	smaller sets rank higher	PS	policy v1	50/50 + heavy100	p99 213 / max 292	✓
O13	matches $\subseteq$ candidates	validator	decision invariant	filter-only	validator subset check	●
O14	TAB/no-strip/no-space	validator	writer	exact-format	validator FAIL on breach	■
O15	list content rules	validator	writer	S2/S3 only, no dup	validator checks	■
O16	PASS gate	validator	<code>validate_submission.py</code>	run gate	manual PASS	✓ (tool)
O17	macro- $F_{0.5}$ formula	PS	<code>evaluate.py</code> (frozen)	per-S1→macro	spec → 0.714	✓
O18	edge cases (1.0/0.0)	PS	<code>evaluate.py</code>	hard-coded	spec cases	✓
O19	recall $\neq$ score	PS	4 eval layers	separated	0.878 vs 0.9486 vs L4	✓ (layers)
O20	$\leq 8B$, MIT/Apache, no LLM	PS	GBM plan (\$16)	LightGBM/XGBoost	model card at selection	■
O21	scale to M rows / 100s M pairs	PS	out-of-core, 16 shards	DuckDB spill	RSS ≈ 648.5 MiB, spill 0	✓
O22	no external lookup	PS	stdlib normalize; label_use=false	no network	manifest assertions	✓
O23	reproducible from code/	PS	manifests, seeds, pins	SHA-256 chain	freeze gate 0/0/0	✓

Legend: ✓ built+verified · ● partially built (contract/artifact present, computation pending) · ■ planned. PS = problem statement; validator = `validate_submission.py`.

31. Current Project Status

Status dashboard from **actual repo evidence**. Nothing planned is presented as implemented (§36).

Component	Status	Evidence	Next Gate
Environment / deps	✓ Existing	<code>.venv</code> 3.12, pinned <code>requirements.txt</code>	—
Ingest + integrity	✓ Existing	<code>integrity_report.md</code> (0 violations)	—
Normalization (frozen)	✓ Existing	<code>normalize.py</code> SHA <code>b3508b60...</code>	immutable
Evaluator (frozen)	✓ Existing	<code>evaluate.py</code> SHA <code>e99d7f0a...</code> → 0.714	immutable
Baseline blocking	✓ Existing	<code>baseline_blocking.md</code> (0.4702 / 0.5061)	—
Four-pass retrieval + IDF rank	✓ Existing	<code>candidate_experiment_summary.md</code> (0.8614 interm.)	—
Candidate policy v1 (FROZEN)	✓ Existing	34,568,979 rows, recall 0.878084 , gate 0/0/0	immutable
Candidate oracle ceiling	✓ Existing	<code>candidate_oracle_report.md</code> 0.948629	—
Matcher split + firewall	✓ Existing	<code>matcher_split_checksums.json</code> , overlap 0/0	protect <code>model_final_eval</code>
Feature spec (65-d)	● In progress	pilot matrix computed (1,098,300 rows, 0/0/0); full-scale running	finish materialization (Phase 8)
Matcher (<code>matcher/</code>)	● In progress	pilot fits (det/LR/GBM-smoke) + <code>lightgbm_smoke_v1.txt</code> ; no search	GBM bake-off (Phase 10)
Negative sampling	● In progress	3 policies run (<code>negative_samples/*.parquet</code>); <code>hybrid_source_balanced</code> selected	full-split sampling (Phase 9)
Calibration + τ	■ Planned	(§17)	tune on calibration (Phase 11)
S1 decision policy	■ Planned	(§17)	build (Phase 12)
France / LOCO probes	■ Planned	(§25)	probe (Phase 13)
Full-scale test inference	■ Planned	memory risk (§19/§27)	measure at scale (Phase 14)
Submission assembler	■ Planned	<code>output/</code> empty	emit two TSVs (Phase 15)
Validator	✓ Existing (tool)	<code>utils/validate_submission.py</code>	run → PASS (Phase 16)

One-line state. The **entire label-free retrieval half is built, frozen, and reproducible** (recall **0.878084**, ceiling **0.9486**). The supervised half has been **piloted end-to-end at smoke scale** (features 1,098,300 rows, negatives 3 policies, matchers det/LR/GBM-smoke fit) and **full-scale feature materialization is running; full-scale training, calibration, decision policy, test inference, and the submission assembler are not built** and `output/` is empty. No end-to-end macro-F_{0.5} (L4) exists yet, and none is claimed; `model_final_eval` is firewalled/untouched.

Open discrepancies carried forward (per §36): (1) **Pilot-vs-planned status (NEW, corrected in this revision):** the earlier draft labelled features/negatives/matcher as "no matrix / no `.fit` / planned", but the repository shows a **completed pilot/smoke end-to-end** (`feature_pilot_*`, `negative_samples/`, `baseline_results.json`, `lightgbm_smoke.json`, `lightgbm_smoke_v1.txt`) — status labels were corrected to ●, and the pilot numbers are explicitly **not** leaderboard/L4 (§14/§15/§16). (2) candidate recall **0.8614 intermediate vs 0.878084 frozen-**

final — reconciled as different objects (§11); (3) **memory budget 0.8 GB vs measured peaks 648.5 MiB / 1.27 GiB / <1.1 GiB** — distinct operations, disk-spill dependent, flagged as inference risk (§19/§27); (4) prior **"137 MB"** → **~600–680 MiB** correction and **leakage-scope** re-scoping logged (§13/§21).

32. Future Extensions

All **[FUT]**; none authorizes editing a frozen stage. Each is an experiment to run and log (§21), kept only on held-out evidence.

- **Candidate policy v2 (new artifact, not an edit) [FUT]**. The oracle brackets show a recall-leaning **75/75** ceiling of **0.952327** and a compact **37/38** at **0.946593** (§12). A v2 could target the either-address-missing cliff (recall **0.729**, §11) — e.g., a phonetic/name-only pass for address-missing S1 — and be frozen as its **own** policy for A/B, leaving v1 intact (ADR-002).
- **France zero-shot hardening [FUT/ORG O9]**. LOCO probes (train {US} eval {India} and vice-versa) to quantify the transfer gap; optionally a deterministic-rule fallback for the France slice if the GBM under-generalizes.
- **Blocking recall recovery [FUT]**. Targeted passes for the **93,134** unreached links (translit-heavy India names, domain-as-name) — measured against candidate-size cost (O12) before adoption.
- **Matcher upgrades [FUT]**. Monotonic constraints on similarity features; per-pair uncertainty for a smarter decision rule; feature ablation to prune the 65-d set.
- **Decision-policy refinement [FUT]**. Per-entity adaptive τ or a light multi-match consistency check — only if it beats a single global τ on `model_calibration`.
- **Inference memory validation [FUT/ENG]**. Measure full-scale test inference RSS against the 0.8 GB budget (projected <1.1 GiB, §19) and tune shard count/batch before the final run.
- **Tracking [FUT]**. Revisit MLflow only if the team scales (ADR-011); not needed now.

Guardrail on all extensions [§36]. Any change that would alter a frozen artifact ships as a **new, separately-frozen** version with its own manifest and A/B evidence — never as an in-place edit to `normalize.py`, `evaluate.py`, the splits, or candidate policy v1.

33. Final Architecture Summary

This architecture resolves business entities across three noisy sources under a **precision-heavy macro-F_{0.5}** objective, with **S1 as the deduplicated reference** and **one-to-many, cross-source** matching as the norm (mean 3.46/S1, 5.6% singletons). It is a disciplined **retrieve-then-score** pipeline with three separated concerns behind hard boundaries: **frozen label-free candidate generation** → **pairwise tabular matcher** → **S1-level decision policy**, feeding the two graded outputs (`matching_results.tsv` , `candidate_pairs.tsv`) through the official validator.

What is real today (✓). Deterministic frozen **normalization** (`b3508b60...`); the frozen **evaluator** reproducing the spec's **0.714**; the leakage-safe **matcher split** (overlaps 0/0); the **FROZEN candidate policy v1** — **34,568,979** candidates, pair recall **0.878084**, p99 **213** / max **292**, reproduced **0/0/0** (checksum `221fef24...9465b067`); the candidate **oracle ceiling 0.948629**; and a **65-feature contract**. The whole label-free retrieval half is built and reproducible.

What has been piloted (●) and what is not yet built (■). A **pilot/smoke end-to-end has run**: 65-feature matrix computed on 1,098,300 pilot candidates (0/0/0), three checksummed negative-sampling policies (`hybrid_source_balanced` selected), and Deterministic/Logistic/LightGBM-smoke matchers fit and thresholded (pilot macro-F_{0.5} 0.629/0.781/0.852); full-scale feature materialization is running. **Not yet built**: full-scale GBM

bake-off/hyperparameter search, calibration/threshold on the full split, the **decision policy, France/LOCO** probes, **full-scale test inference**, and the **submission assembler** — `output/` is **empty**. `model_final_eval` is **firewalled/untouched** — **no end-to-end score (L4) exists and none is claimed**.

Design commitments that make it competition-safe. Precision-first thresholding and a first-class singleton path for the $F_{0.5}$ regime; per-candidate keep/drop (never top-1) for one-to-many; **country-as-agreement** for France zero-shot; **out-of-core** execution (16 shards + DuckDB spill) for the 0.8 GB budget; **fair-play** by construction (`label_use=false`, stdlib-only normalize, no network); and **reproducibility** via a SHA-256 manifest chain plus an append-only ledger.

Honesty ledger (surfaced, not hidden). Recall **0.8614** (intermediate) vs **0.878084** (frozen-final) are different objects; the **0.8 GB** budget is a resident target that measured peaks (**648.5 MiB** / $\approx$ **1.27 GiB** audit / **<1.1 GiB** projected inference) can exceed via disk spill — a flagged inference risk; and two prior claims ("**137 MB**", "**no leakage by construction**") were corrected and logged.

Bottom line. The retrieval foundation is frozen, measured, and reproducible with a **0.9486** score ceiling; the remaining work is the supervised scoring-and-decision half, whose contracts, splits, and guardrails are already fixed so it can be built without disturbing anything frozen upstream.

End of master architecture. Companion artifacts: `amazon_ml_challenge_architecture_diagrams.md` (all Mermaid diagrams), `amazon_ml_challenge_master_architecture.pdf` (polished render).